

Hesperidin contributes to the vascular protective effects of orange juice: a randomized crossover study in healthy volunteers.



BACKGROUND: Although numerous human studies have shown consistent effects of some polyphenol-rich foods on several intermediate markers for cardiovascular diseases, it is still unknown whether their action could be specifically related to polyphenols. **OBJECTIVE:** We investigated the effect of orange juice and its major flavonoid, hesperidin, on microvascular reactivity, blood pressure, and cardiovascular risk biomarkers through both postprandial and chronic intervention studies. **DESIGN:** Twenty-four healthy, overweight men (age 50-65 y) were included in a randomized, controlled, crossover study. Throughout the three 4-wk periods, volunteers daily consumed 500 mL orange juice, 500 mL control drink plus hesperidin (CDH), or 500 mL control drink plus placebo (CDP). All measurements and blood collections were performed in overnight-fasted subjects before and after the 4-wk treatment periods. The postprandial study was conducted at the beginning of each experimental period. **RESULTS:** Diastolic blood pressure (DBP) was significantly lower after 4 wk consumption of orange juice or CDH than after consumption of CDP ($P = 0.02$), whereas microvascular endothelium-related reactivity was not significantly affected when measured after an overnight fast. However, both orange juice and CDH ingestion significantly improved postprandial microvascular endothelial reactivity compared with CDP ($P < 0.05$) when measured at the peak of plasma hesperetin concentration. **CONCLUSIONS:** In healthy, middle-aged, moderately overweight men, orange juice decreases DBP when regularly consumed and postprandially increases endothelium-dependent microvascular reactivity. Our study suggests that hesperidin could be causally linked to the beneficial effect of orange juice. This trial is registered at clinicaltrials.gov as NCT00983086.